

Unit1. Vocabulary

Term	Definition
Science	Systematic study of the natural world through observation and experiment.
Technology	Application of scientific knowledge to create solutions for human needs.
Technoscience	Interconnected combination of scientific research and technological development.
Scientific Method	Systematic methodology used to gain objective knowledge about the universe.
Observation	Attentively examining natural phenomena.
Hypothesis	Proposing a reasoned explanation for observed phenomena.
Experimentation	Designing tests and measuring variables to validate or reject a hypothesis.
Conclusion	Determining if experimental data supports the hypothesis.
Knowledge ("To Know")	Theoretical understanding of scientific laws and models.
Execution ("To Make")	Practical action of building products, tools, and processes.
Technological Process	Ordered phases to turn a practical problem into a technical solution.
Design Constraints	Project limits or requirements such as budget, time, materials, or safety.
Problem Description	Precise formulation of the need or challenge to solve.
Information Research	Exploring existing solutions, literature, and consulting experts.
Brainstorming	Group creativity technique to generate many ideas without early judgment.
Mind Mapping	Visual diagram used to organize connected ideas around a central theme.
Sketch	Quick, freehand drawing of an idea without exact measurements.
Bill of Materials	Table listing required components, tools, materials, and estimated costs.
Process Sheet	Step-by-step plan detailing tasks, team members, timing, and safety.
Prototype	Early working model built to test and validate a design.
Assembly	Fitting components together according to design specifications.
Testing	Evaluating if the constructed object operates safely and meets requirements.
Redesign	Making adjustments and improvements based on test results.